



Original research

Kinematic and kinetic comparison between American and Japanese collegiate pitchers



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ABSTRACT

Objectives: Understanding the differences in baseball pitching biomechanics between American and Japanese pitchers may help with training and developing these athletes. The purpose of this study was to investigate the kinematic and kinetic differences in collegiate baseball pitchers from United States of American and Japan.

Design: Controlled laboratory study.

Methods: Data were analyzed for 11 American and 11 Japanese collegiate pitchers throwing fastballs using 3D motion capture (480 Hz).

Results: The Americans were heavier (95 ± 7 kg vs 81 ± 7 kg), taller (189 ± 3 cm vs 180 ± 6 cm), and had faster ball velocity (39 ± 1 m/s vs 35 ± 2 m/s). By the end of arm cocking phase, the American pitchers had rotated their shoulder to a greater degree ($p = 0.021$, $d = 1.5$) and at ball release the Japanese had greater knee flexion ($p = 0.020$, $d = 1.2$). American pitchers exhibited greater peak kinetics on the throwing arm; however, when normalized for height and weight only three differences remained.

Conclusion: The differences found between the American and Japanese players could contribute to the increased ball velocity in the American pitchers. Additionally, throwing arm peak kinetics were greater in the American pitchers which may help generate greater ball velocity; however, increased kinetics may also lead to increased risk of injury.

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Practical implications

- American pitchers were heavier, taller, had greater kinetics, and pitched with faster ball velocity than the Japanese pitchers.
- Understanding the differences between countries offers coaches and trainers a more effective approach in effectively training pitchers.

1. Introduction

Baseball is returning to the Olympics since its brief hiatus in 2008. The 2020 Summer Games will be held in Tokyo, Japan, where the sport is wildly popular. Two countries with perhaps the highest level of play, based upon success in international competition and salaries in professional leagues, are the United States and Japan.

There are significant cultural differences in their approach to the game such as game schedule, style, and construction of the mound and the ball.¹ Coaches who have trained players globally have speculated differences in coaching instruction result in different styles of pitching.² Understanding biomechanical differences between cultures could lead to developing strategies for winning in international competition.

Although there is research from different countries on pitching biomechanics, only two investigations have specifically compared American and Japanese pitchers. In the first study, Escamilla et al.³ manually digitized video (120 Hz) during the 1996 Olympics and compared pitchers from nine different countries, examining kinematics only. At foot contact (FC), Japanese pitchers threw with less shoulder horizontal abduction and less shoulder external rotation (ER) compared to the other international pitchers. A second study examined differences between American and Japanese professional pitchers using three-dimensional motion capture (240 Hz).⁴ The authors used data from two different labs and reported Japanese pitchers exhibited greater pelvis rotation and elbow flexion at FC,

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Table 1
Measures for Japanese and American collegiate pitchers (M ± SD).

	American (n = 11)	Japanese (n = 11)	p value	Cohen's d
Age (years)	21 ± 1	21 ± 1	0.582	0.2
Height (cm)	189 ± 3	180 ± 6	0.004*	1.9 ^j
Mass (kg)	95 ± 7	81 ± 7	0.005*	1.8 ^j
Arm length (cm)	58 ± 3	52 ± 4	0.002*	1.7 ^j
Arm length (% of height)	31 ± 2	29 ± 2	0.026*	1.0 ⁱ
<i>Instant of Foot contact</i>				
Stride length (% body height)	81 ± 3	82 ± 6	0.730	0.2
Elbow flexion (°)	97 ± 21	116 ± 16	0.058	1.0 ⁱ
Shoulder abduction (°)	83 ± 10	83 ± 10	0.900	0.1
Shoulder horizontal abduction (°)	38 ± 9	29 ± 16	0.163	0.7
Shoulder external rotation (°)	41 ± 31	52 ± 32	0.389	0.4
Lead knee flexion (°)	48 ± 7	51 ± 7	0.489	0.3
Pelvis rotation (°)	54 ± 14	59 ± 14	0.431	0.4
Trunk rotation (°)	42 ± 12	43 ± 9	0.700	0.1
<i>Instant of Maximum shoulder rotation</i>				
Elbow flexion (°)	86 ± 7	90 ± 12	0.444	0.4
Shoulder abduction (°)	90 ± 10	98 ± 10	0.111	0.8
Shoulder external rotation (°)	167 ± 6	159 ± 4	0.021*	1.5 ⁱ
Trunk flexion (°)	−5 ± 9	2 ± 11	0.099	0.7
<i>Instant of Ball release</i>				
Elbow flexion (°)	29 ± 7	25 ± 7	0.275	0.6
Shoulder abduction (°)	92 ± 9	99 ± 5	0.052	1.0 ^j
Lead knee flexion (°)	29 ± 15	45 ± 12	0.020*	1.2 ^j
Trunk flexion (°)	14 ± 7	20 ± 11	0.152	0.7
Trunk lateral flexion (°)	−29 ± 6	−27 ± 8	0.695	0.2
Ball velocity (m/s)	39 ± 1	35 ± 2	<0.001*	2.1
<i>Peak Angular Velocities and Energies</i>				
Maximum knee extension velocity (°/s)	390 ± 149	265 ± 120	0.078	0.9 ^j
Maximum pelvis rotational velocity (°/s)	653 ± 64	612 ± 63	0.147	0.7
Maximum trunk rotational velocity (°/s)	667 ± 156	709 ± 167	0.563	0.3
Maximum shoulder internal rotation (°/s)	6556 ± 1391	6914 ± 2004	0.689	0.2
Maximum elbow extension velocity (°/s)	2419 ± 325	2435 ± 363	0.904	0.0
Pelvis rotational kinetic energy (J)	33 ± 9	31 ± 11	0.672	0.2
Trunk rotational kinetic energy (J)	39 ± 7	35 ± 7	0.098	0.6

Note: Data are presented as mean (± standard deviation). * $p < 0.05$. ^j Cohen's d over 0.80 denoting a large effect size. Shoulder horizontal abduction/adduction of 0° was used as a reference point. Negative adduction value was considered abduction. Pelvis rotation of 0° was used as the anterior pelvis perpendicular to the batter, negative value was considered rotation past the batter. Negative trunk flexion value was considered trunk extension and positive trunk flexion value was considered flexion. Negative trunk lateral flexion value was considered leaning toward the non-throwing arm and positive values were considered leaning towards the throwing arm.

elbow flexion at maximum ER, and greater shoulder abduction at ball release (BR). Additionally, American pitchers pitched with faster ball velocity and had greater elbow varus torque.⁴ The setwo studies revealed differences in pitching biomechanics between professional American and Japanese pitchers; however, additional data in collegiate players are needed to contribute to the understanding of this topic.

The purpose of this study was to investigate pitching between American and Japanese collegiate baseball pitchers using 3D motion capture (480 Hz). Specifically, we compared full body mechanics of a pitch at the time points of FC, maximum ER, and BR as well as peak throwing arm kinetics between the two groups. We hypothesized the Americans would have a less elbow flexion at FC, greater shoulder ER at maximum ER, and faster ball velocity compared to the Japanese pitchers. Additionally, we hypothesized American pitchers would have increased throwing arm kinetics compared to Japanese players associated with higher ball velocity.

2. Methods

A sample of 22 collegiate pitchers (11 American and 11 Japanese) were included in this study (Table 1). The American pitchers were born and raised in the United States and currently pitched for an American college. The Japanese pitchers were born and raised in Japan and pitched for a Japanese college team that was training in the United States at the time of the study. Pitchers were considered

healthy if they were able to perform at 100% capacity⁵ and had no history of moderate to severe injury requiring rest/rehabilitation longer than two weeks. This study was approved by the Sterling Institutional Review Board (Atlanta, GA, USA).

Participants were instrumented with 38 reflective markers on anatomical landmarks, including: bilateral placement on the second metatarsal, posterior calcaneus, lateral and medial malleolus, lateral and medial femoral epicondyle, greater trochanter, anterior and superior iliac spine, lateral tip of the acromion, medial clavicle, lateral and medial humeral epicondyle, radial styloid, and ulnar styloid. Additional markers were placed on the T2, T8, xyphoid process, and on the third metacarpal of the throwing hand. Participants were then allowed as much time as needed to perform a warm-up routine of choice.

When ready, participants pitched off a dirt mound, constructed in America, to a catcher at home plate regulation distance away (18.4 m). Each pitcher threw 8–10 fastball with game-like effort. Ball velocity was recorded with a radar gun (Stalker Sports Radar, Plano TX). The five pitches with the fastest ball velocity, regardless of pitch outcome, for each player were used in the analysis.⁶

Position coordinate data of the reflective markers were collected using motion analysis system at 480 Hz (Motion Analysis Corp., Santa Rosa, CA). Marker data were transformed to the ISB global coordinate standard: Y was vertically upward, X direction was perpendicular to Y and positive when pointed towards home plate, and Z was the cross product of X and Y.

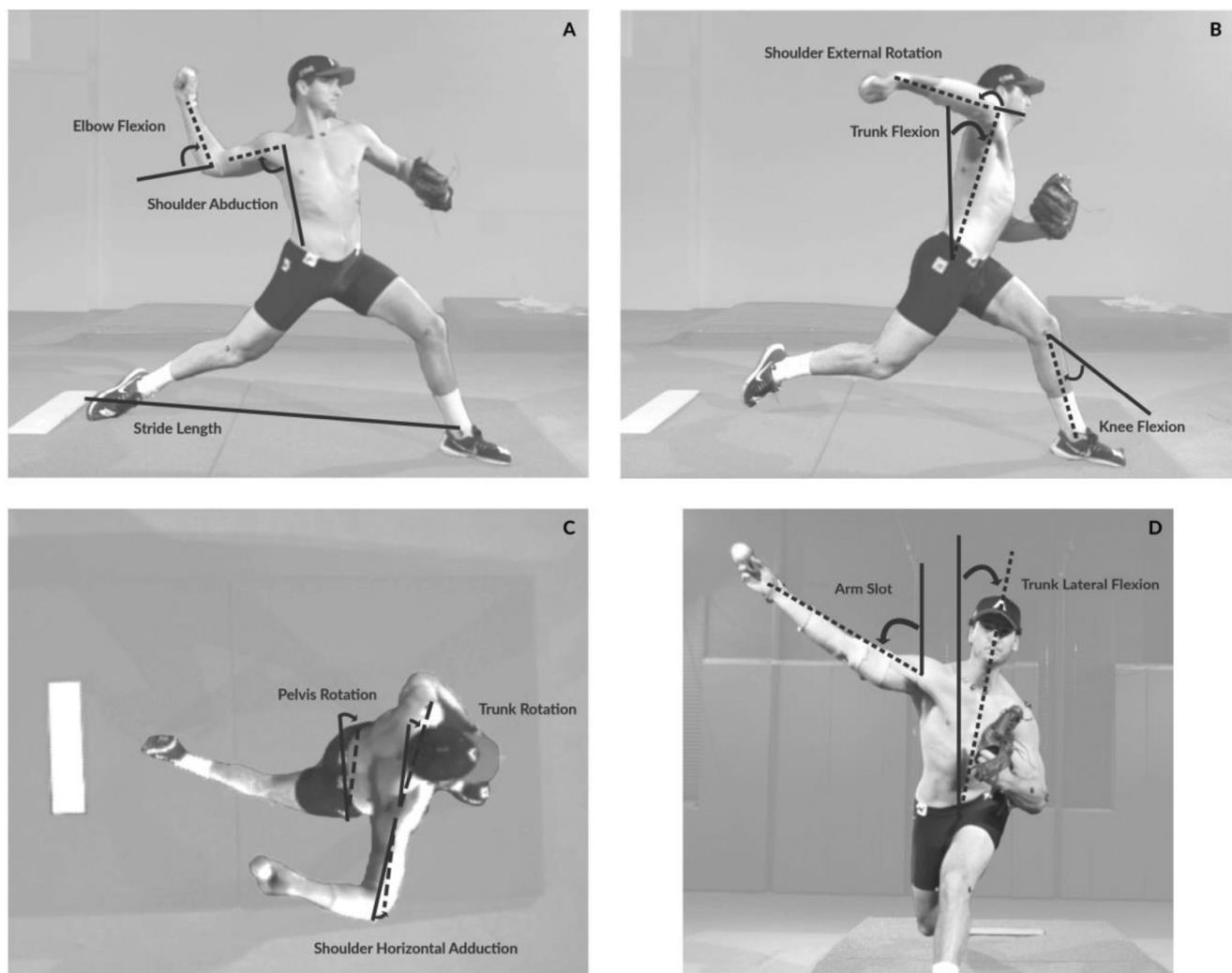


Fig. 1. Definitions of kinematic variables for A) elbow flexion, shoulder abduction, and stride length, B) shoulder external rotation, upper trunk flexion, and knee flexion, C) upper trunk rotation, pelvis rotation, and shoulder horizontal adduction, and D) arm slot and upper trunk lateral flexion.

Data processing was performed using custom MATLAB code (The MathWorks, Natick, MA). The human body was modelled as a system of 14 articulated rigid segments (feet, shanks, thighs, pelvis, trunk, upper arms, forearms, and throwing hand). Marker data were lowpass filtered (fourth-order, zero lag Butterworth filter, cut-off frequency of 13.4 Hz)⁷ and used to define segment and joint coordinate systems according to ISB recommendations.^{8,9} Dynamics of the segments superior to and including the pelvis were computed using Newton-Euler equations and used to estimate net muscle moments and net intersegmental forces for each joint.¹⁰ Normalized kinetics were calculated using body weight (N) and height (m). Joint angles were determined according to ISB recommendations^{8,9} with estimates of knee joint angles modified as in Dabirrahmani and Hogg.¹¹ Joint angles were once differentiated to estimate joint angle velocities. Rotational kinetic energy of segment k (T_k) was computed using the following equation, where ω_i and I_i are the angular velocity and inertia tensor of segment k .

$$T_k = \frac{1}{2} \omega_k^T I_k \omega_k$$

Joint kinematics were used to identify the instants of FC, maximum shoulder ER, and BR. FC was defined as the first frame that either the lead toe or heel reached minimum Y, whichever came first. Maximum shoulder ER was established as the frame in which the throwing arm reached maximum ER. BR was identified as the

instant 0.01 s after the wrist marker passed the elbow marker in the X direction.¹²

Elbow flexion was the angle between the forearm and upper arm (Fig. 1a). Knee flexion was the angle between the thigh and shank (Fig. 1b). Both elbow and knee flexion angles were 0° at full extension. Shoulder abduction was the angle between the distal direction of the upper arm and the line between the two shoulders with respect to the frontal plane (Fig. 1a). Shoulder external rotation was the longitudinal rotation of the upper arm relative to the upper trunk's sagittal plane, so that 90° was when the humerus was parallel to the trunk (Fig. 1b). Shoulder horizontal abduction was the angle between the upper throwing arm and the upper trunk vector in the upper trunk's transverse plane, such that a negative value was abduction (Fig. 1c). Pelvis rotation was the angle of the pelvis line (a line connecting the right and left hip) to the pitch direction; defined as 90° when the pelvis was aligned with the global X direction (Fig. 1c). Trunk rotation was the angle between the pelvis and upper trunk in the transverse plane (Fig. 1c). Trunk lateral flexion was the angle between the pelvis and upper trunk in the coronal plane and defined as 0° when the upper trunk line was parallel to the pelvis line, and positive value indicated tilted toward the glove hand (Fig. 1d). Trunk flexion was the angle between the superior direction of the upper trunk compared to the pelvis in the sagittal plane, where a positive angle reflected forward upper trunk flexion

Table 2

Peak kinetic variables and differences between American and Japanese collegiate pitchers (M ± SD).

	American	Japanese	p value	Cohen's <i>d</i>
<i>Arm cocking</i>				
Shoulder internal rotation torque (Nm)	93 ± 10	68 ± 11	<0.000*	2.3 [†]
Normalized (%BW.BH)	5.3 ± 0.6	4.7 ± 0.7	0.054	0.9 [†]
Shoulder horizontal adduction torque (Nm)	100 ± 10	73 ± 20	0.007*	1.7 [†]
Normalized (%BW.BH)	5.7 ± 0.4	5.1 ± 1.2	0.160	0.7
Shoulder superior force (N)	210 ± 81	202 ± 60	0.819	0.1
Normalized (%BW)	20.9 ± 10.9	25.3 ± 7.5	0.314	0.5
Elbow varus torque (Nm)	90 ± 9	71 ± 14	0.003*	1.6 [†]
Normalized (%BW.BH)	5.2 ± 0.6	5.0 ± 0.9	0.460	0.3
Elbow medial force (N)	379 ± 56	353 ± 71	0.270	0.4
Normalized (%BW)	40.9 ± 6.3	44.1 ± 7.0	0.098	0.5
<i>Arm acceleration</i>				
Shoulder anterior force (N)	276 ± 115	308 ± 100	0.395	0.3
Normalized (%BW)	29.9 ± 12.7	38.6 ± 11.8	0.063	0.7
Elbow anterior force (N)	366 ± 41	279 ± 61	0.000*	1.7 [†]
Normalized (%BW)	39.6 ± 4.9	34.7 ± 5.4	0.003*	1.0 [†]
Elbow flexion torque (Nm)	66 ± 8	42 ± 9	0.000*	2.8 [†]
Normalized (%BW.BH)	3.8 ± 0.4	2.9 ± 0.4	<0.000*	2.1 [†]
<i>Arm deceleration</i>				
Shoulder adduction torque (Nm)	139 ± 36	81 ± 35	0.005*	1.6 [†]
Normalized (%BW.BH)	8.0 ± 2.0	5.7 ± 2.4	0.035*	1.0 [†]
Shoulder distractive force (N)	1064 ± 96	951 ± 197	0.192	0.7
Normalized (%BW)	114.7 ± 7.2	118.6 ± 20.0	0.621	0.3
Elbow distractive force (N)	1065 ± 124	899 ± 166	0.068	1.1 [†]
Normalized (%BW)	114.8 ± 10.7	112.3 ± 17.5	0.763	0.2

Note: Data are presented as mean (± standard deviation). * $p < 0.05$. [†] Cohen's *d* over 0.80 denoting a large effect size. Data were normalized by body weight (N) for force and body weight (N) and height (m) for torques.

(Fig. 1b). Stride length was calculated as the distance between the back and the lead ankle at FC, as a percentage of height (Fig. 1a). Arm length was calculated as the distance between the acromion marker and the ulnar styloid marker, expressed in both cm and as a percentage of height.

Each dependent variable was averaged across the five pitches with the fastest ball velocity for each pitcher and were used in this analysis. Five pitches were considered acceptable for characterizing each participant, as previous research reported low within-subject variability in collegiate pitchers.¹³ Unpaired *t*-tests were used to assess differences in variables between American and Japanese pitchers. The alpha level was set at 0.05.⁴ Cohen's *d* was calculated to assess the effect size for each independent samples *t*-test. When necessary, values of 0.2, 0.5, and 0.8 were used to characterize small, medium, and large effect sizes, respectively.¹⁴

3. Results

American pitchers were taller ($p = 0.004$, $d = 1.9$) and heavier ($p = 0.005$, $d = 1.8$) than Japanese pitchers; but there were no differences in age between the two groups ($p = 0.581$, $d = .20$) (Table 1). Americans had longer arm length by 6 cm ($p < 0.001$, $d = 1.7$) and faster ball velocity ($p < 0.001$, $d = 2.1$) compared to the Japanese.

There were only two kinematic variables that were significantly different between the two groups (Table 1). At the instant of maximum ER, American pitchers exhibited greater shoulder ER ($p < 0.021$, $d = 1.5$). At BR, Japanese pitchers displayed greater lead knee flexion ($p < 0.020$, $d = 1.2$). There were no differences in peak angular joint and segment velocities between groups.

American pitchers displayed greater kinetics compared to their Japanese peers (Table 2). Of the 11 absolute kinetic variables reported, 6 were statistically greater in the American pitchers than the Japanese: shoulder internal rotation torque ($p < 0.000$, $d = 2.3$), shoulder horizontal adduction torque ($p = 0.007$, $d = 1.7$), elbow varus torque ($p = 0.003$, $d = 1.6$), elbow anterior force ($p = 0.000$, $d = 1.7$), elbow flexion torque ($p = 0.000$, $d = 2.8$), shoulder adduction torque ($p = 0.005$, $d = 1.6$).

When the kinetics were normalized by height and weight, only three remained statistically different (Table 2): elbow anterior force ($p = 0.000$, $d = 1.0$), elbow flexion torque ($p < 0.000$, $d = 2.1$), shoulder adduction torque ($p = 0.035$, $d = 1.0$).

4. Discussion

The hypothesis of this investigation was confirmed in that there were significant differences between Japanese and American pitchers for pitching kinematics. The differences in pitching biomechanics highlight different mechanical strategies employed by the two groups which may have contributed to differences in ball velocity. The magnitude (35–39 m/s) and difference (10% between American and Japanese) of ball velocities reported in the current investigation are similar to those previously reported on international pitching.¹⁵ It has been shown that high velocity pitchers tend to be taller and heavier^{15,16} and may help explain the increased ball velocity for the American pitchers in the current study. However, differences in the pitching kinematics and kinetics are also likely to have contributed to the increased velocity by the American pitchers.

Establishing the proper rotational base of support is essential for a pitcher to have sequential rotation of body segments and an efficient delivery. Both groups fell within previously reported ranges for stride length of collegiate and professional baseball pitchers of 80–87 % of height.^{4,12,17} Stride length in this study was on the low end of this range, likely due to the difference in calculation of stride length; previous studies utilized the centre of the pitching rubber as measure^{3,15,18} or the difference between ankle joint centres at FC.¹⁷ Regardless of measurement calculation, we reported no statistical difference between the two groups whereas the two previous studies have reported Japanese pitchers had a longer stride length.^{3,4} It is possible that differences in mound stiffness between the two countries¹⁹ could influence pitch stride length, as some Japanese pitching mounds still use softer dirt than American pitching mounds. In the current study, all pitchers threw off an American made mound and this could be masking potential differences. How-

ever, no research to date has been conducted to investigate these effects.

Oi et al.⁴ reported no differences between Japanese and American pitchers for lead knee flexion at both FC and BR. Lead knee extension range of motion has been reported greater in professional American pitchers compared to collegiate pitchers,^{3,16} implying more skilled players utilize knee extension during a pitch. Greater lead knee extension is thought to drive the pelvis to rotationally accelerate forward and aid in the transfer of energy to the upper trunk, resulting in greater energy transfer available for the throwing arm and ultimately ball velocity. High velocity pitchers have demonstrated greater total lead knee 'excursion' (defined as the change in knee flexion from FC to BR)(19°) compared to low velocity pitchers (2°).¹⁷ Similar findings were reported in American pitchers, the high velocity pitching group extended their lead knee through a greater range and at a faster velocity compared to the low velocity pitching group.¹⁶ In the current study, the American pitchers presented with greater lead knee excursion (19° compared to 6° for the Japanese pitchers), suggesting they had increased contribution from the lead leg in generating and transferring energy through the kinetic chain.

In theory, greater lead knee excursion and angular knee extension velocity positively impacts the kinetic chain resulting in greater pelvis and trunk angular velocity. Surprisingly, in this study there was no statistical difference in peak pelvis and trunk angular velocities. Previous investigations of international pitchers have reported varied findings, establishing either no differences in segment angular velocities,^{4,16} similar segment angular velocities,¹⁵ as well as patterns that report both faster pelvis and trunk angular velocities for high velocity groups compared to low velocity groups.¹⁷ It is possible that although sequential rotation of body segments is important for energy transfer, angular velocity alone, which does not consider the mass of the player, may not be ideal for determining contribution to overall energy transfer. A more useful relationship might be rotational kinetic energy (KE), or the energy due to rotation. In the current study there were no statistical differences between groups for both pelvis and trunk rotational energy. Only one other study has investigated rotational kinetic energy in pitchers and found no difference between high velocity and low velocity professional pitchers.²⁰ In the current study, trunk KE was reported 35–39 J which was similar to Luera et al.'s²⁰ values of 36 J. However, our values were less for pelvis KE (31–33 J) compared to the aforementioned study (36–39 J). Research needs to be conducted on this principle to fully understand the contribution of rotational energy to performance in pitching.

Shoulder ER at FC has been reported in literature as highly variable, ranging from 35° to 106°.^{3,12,21} The current study found no differences in ER at FC between the American and Japanese pitchers; however, previous studies on international pitchers have reported Japanese pitchers to have shoulder ER both lower³ and higher²¹ than American pitchers. It is possible that differences in how the FC frame was determined caused differences between the reported measures across the different studies. In the current study, FC was defined as the first instance the toe or heel marker landed on the mound while other studies identified FC as when the entire foot was placed on the mound. In this study, American pitchers externally rotated the shoulder through a greater range during the arm cocking phase (126° compared to 107°), which is similar to the differences noted between Korean and American pitchers.¹⁵ It has been suggested that increased range of motion in the shoulder during a pitch is associated with increased ball velocity due to the stretch-shortening cycle in the shoulder muscles.²² Theoretically, the greater the arm externally rotates, the longer the distance a pitcher has to apply a force to the ball; thus resulting in increased ball velocity.²³ Matsuo et al.¹⁶ reported greater maximum shoulder ER was associated with higher throwing velocities in American

professional and collegiate pitchers. Conversely, it has been shown that shoulder maximum ER was not associated with ball velocity.²⁴ An important factor to consider is that while increased ER could be associated with increased ball velocity it has also been associated to increased kinetics and injury.^{25,26} In this study, the American pitchers displayed statistically significantly greater shoulder kinetics during arm cocking which may be related to increased shoulder ER and horizontal adduction kinematics.

Six of the eleven measured absolute kinetic values were greater in the American group: shoulder internal rotation torque, shoulder horizontal adduction torque, elbow varus torque, elbow anterior force, elbow flexion torque, and shoulder adduction torque. However, as pointed out by Oi et al.,⁴ it is unclear if normalized kinetics are more applicable when comparing groups of different height and weight as there is no research to indicate if there exists size and strength differences in joint ligaments and tendons. When the kinetics were normalized, the differences remained for only three: elbow anterior force, elbow flexion torque, and shoulder adduction torque. Interestingly, Oi et al.⁴ reported the only difference for normalized kinetics was 1% BW.BH greater shoulder horizontal adduction torque in Japanese pitchers.

There were limitations to this study that warrant mentioning. While this study captured data from pitchers throwing from a dirt mound in an outdoor bullpen, the pitchers wore only spandex which is atypical for practice and game. Pitchers were instructed to throw with full-effort but it is possible that because testing was not a natural environment, they pitched with reduced effort. Additionally, pitchers were asked to throw only fastballs and the differences between American and Japanese players might not be applicable to other pitch types. Because of the small sample size, it is possible that differences were masked and a larger sample would reveal significant differences. There was a possibility for Type 1 error due to multiple t-tests performed; however, we are confident in the conclusions of this study.

5. Conclusion

In this study, American and Japanese pitching biomechanics were compared and revealed differences that may affect performance. The Americans had greater maximum shoulder ER and less lead knee flexion at ball release. Shoulder and elbow kinetics were greater in the American pitchers which potentially could help generate greater ball velocity; however, increased kinetics may also lead to increased risk of elbow injury. Understanding differences in athletes developed in these two countries is important for trainers and coaches in mitigating injury risk while maximizing performance.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jsams.2020.04.013>.

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